

Mohammed Suhail Rafeeq V Software Engineer

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PROFESSIONAL EXPERIENCE

Founding Engineer, Cawosh 	10/2025 – Present UK (Remote)
- Designed and built backend services and cloud infrastructure for a multi-tenant garage management platform. - Built a customer-facing quick booking platform enabling users to discover garages and book services online. - Led development within a small engineering team, building scalable services and deployment workflows using Python, AWS, Docker Swarm, and Pulumi. - Architected a cost-effective containerized deployment platform, balancing operational simplicity, reliability, and infrastructure costs.	
Software Engineer, TurboML 	10/2024 – 09/2025 India (Remote)
- Delivered a 10x Python SDK performance improvement by implementing Apache Arrow-based zero-copy ingestion integrated with C++ execution runtimes. - Redesigned snapshot handling and built runtime portability between local in-process and remote C++ execution runtimes, improving reliability, debuggability, and deployment workflows. - Helped scale Puch AI to 1M+ users, supporting high-concurrency messaging workloads. - Designed and implemented Puch AI's initial RAG pipeline using Qdrant for low-latency, contextual retrieval. - Redesigned conversation storage and retrieval, improving context handling and LLM response quality.	
Software Developer, SOTI 	09/2020 – 12/2023 Kochi, India
- As part of the Core Performance Team, contributed to system architecture and scalability improvements supporting 1M+ managed devices. - Designed caching layers for read-heavy backend services, reducing tail latency by 3x and improving reliability at scale. - Contributed to Linux device management agent development in C++, building proof-of-concepts for architectural changes and performance optimizations. - Mentored engineers on .NET CLR internals, C++ runtime systems, and low-level debugging.	

SKILLS

- **Database Systems** — PostgreSQL Internals, Storage Engines, Query Execution, Buffer Management, MVCC, Apache Arrow, DataFusion, Arrow Flight SQL, OLAP Systems
- **Programming Languages** — Rust, C/C++, Go, Python, SQL, C#, TypeScript
- **Distributed Infrastructure & Performance** — Kubernetes, AWS, Linux, Containers, Pulumi, eBPF, Valgrind, Profiling, Performance Tuning, Debugging

PROJECTS

pgFusion  , Native PostgreSQL-to-Arrow OLAP System	2025
- Built a high-performance OLAP system executing analytical queries directly on PostgreSQL heap files without ETL or data duplication. - Developed a native PostgreSQL-to-Apache Arrow reader, integrating Apache DataFusion, Arrow Flight SQL, and predicate/projection pushdown for vectorized query execution.	
NaadanDB  , Experimental Hybrid Relational Database	2024
- Built an experimental relational database exploring storage engine design, MVCC, and query execution for hybrid OLTP/OLAP workloads. - Designed and implemented core database components, including a buffer manager, MVCC-based transaction processing, page-based storage, and query execution primitives.	
CloudBnB  , Distributed Cloud Platform	2020
- Built a distributed cloud platform that aggregates idle machines into a shared compute and storage cluster. - Implemented object storage, serverless function execution, and application deployment capabilities. - Designed distributed services for resource provisioning, workload scheduling, and application lifecycle management.	
Rusty-Fuzzer  , Coverage Guided Evolutionary Fuzzer	2019
- Developed a Rust-based evolutionary fuzzer leveraging LLVM Sanitizer Coverage instrumentation for automated vulnerability discovery.	

EDUCATION

Bachelor of Technology, Computer Science and Engineering , School of Engineering, CUSAT	2016 – 2020 Kochi, India
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